

N.S. LABORATORY SDN. BHD.

HEXINE, 20mg/g, Powder for use in drinking water

Active Ingredient:

Each gram contains

Bromhexine hydrochloride -----20mg

Product Description:

White or almost white powder which can be dissolved in animal drinking water.

Dosage form:

Powder for use in drinking water

Pharmacodynamics:

Bromhexine is a mucoregulator. By activating the secretion of the seromucous glands, bromhexine helps to re-establish the viscosity and elasticity of bronchial secretions in the tracheobronchial tree. In addition, its expectorant action encourages mobilization of mucus and enables effective bronchial drainage, thereby improving the functioning and defence capability of the lung. These two simultaneous actions lead to an abundant discharge and facilitate a productive cough. It breaks down the network of acid glycoprotein fibres found in mucoid sputum, which are mainly responsible for the characteristic viscosity.

Pharmacokinetics:

Absorption- In pigs, bromhexine is rapidly absorbed following oral administration with a peak plasma concentrations obtained in one to three hours. The concentration plateau is reached 12 hours after the second or third administration.

In cattle, plasma concentrations increase progressively over several hours following administration.

In turkeys or chickens, peak plasma concentrations are reached within 2 to 4 hours of oral administration.

Distribution- Due to the lipophilic character of bromhexine, it has a strong affinity for lipid tissues and a slow depletion profile from these tissues.

Metabolism- Bromhexine is largely metabolised into more polar compounds.

Elimination- The apparent half-life of elimination of total plasma radioactivity after the last administration is 20 to 30 hours in a pig, 40 to 50 hours in cattle and 40 to 50 hours in chickens and turkeys.

Indication:

Mucolytic treatment of congested respiratory tract.

Mode of Administration:

Oral administration via drinking water.

Recommended Dosage:

For oral use in drinking water.

0.45 mg of bromhexine per kg bodyweight daily, equivalent to 2.5 g of product per 100 kg bodyweight per day administered for 3 to 10 consecutive days, in drinking water.

The following formula may be used to calculate the required concentration of product

(in milligrams of product per litre drinking water):

$$\frac{25\text{mg of the product}}{\text{per kg bodyweight}} \times \frac{\text{Average bodyweight (kg)}}{\text{of animals to be treated and per day}}$$

$$\frac{\text{Average daily water intake (l/animal)}}{\text{= x mg of the product per litre of drinking water}}$$

The required amount of product should be weighed as accurately as possible using suitably calibrated weighing equipment. The intake of medicated water depends on the clinical condition of the animals. The maximum solubility of the product is 100 g/l in water at 20°C. The time required for complete dissolution varies from 3 minutes (10 g/l) to 15 minutes (100 g/l). For stock solutions and when using a proportioner, take care not to exceed the maximum solubility. Adjust flow rate settings of the dosing pump according to the concentration of the stock solution and the water intake of the animals to be treated. Any unused medicated water should be discarded after 24 hours. Care should be taken that the intended dose will be completely ingested.

Contraindications:

Do not use in cases of pulmonary oedema.

Do not use in cases of hypersensitivity to the active substance or to any of the excipients.

Special Warnings for each Target Species:

None.

Special Precautions for Use in Animals:

In case of serious lungworm infection, the drug should only be used 3 days after the commencement of the anthelmintic treatment.

Special Precautions to be taken by the Person Administering the Veterinary Medicinal Product to Animals:

This product may cause hypersensitivity reactions (allergy). People with known hypersensitivity to bromhexine or lactose should avoid contact with the product. During preparation and administration, inhalation of dust particles should be avoided. Wear an appropriate mask when handling the product. If respiratory symptoms develop following exposure, seek medical advice and show this warning to the physician. This product may cause irritation to the skin, eyes and mucous membranes. Avoid direct contact with the product. Wear gloves and protective glasses during the use of the product. Wash hands and any exposed skin after use. If accidental contact occurs, rinse the affected area with large amounts of clean water. Do not eat, drink or smoke while handling this product.

Interactions with other Medicaments:

The product may be used in conjunction with antibiotics and/or sulphonamides and bronchodilators. Bromhexine modifies the distribution of antibiotics in the organism and increases their concentration in the serum and in the nasal secretions (e.g. spiramycin, tylosin and oxytetracycline). When administered concomitantly with the product, antimicrobial agents should, nevertheless, not be underdosed.

Usage during Pregnancy and Lactation:

Studies in laboratory animals have not produced evidence of foetotoxic effects or effects on fertility at the recommended dose. However this has not been specifically studied in the target species. Use only according to the benefit-risk assessment by the responsible veterinarian.

Adverse Effects:

None known.

Overdose and Treatment:

None known.

Withdrawal Periods (Meat and Offal):

Cattle (calves)- 2 days, Not authorized for use in animals producing milk for human consumption;

Pigs- 0 days;

Chickens, turkeys and ducks- 0 days, Not for use in birds producing eggs for human consumption, during and 4 weeks before the laying period.

Storage Condition:

Store in cool, dark place (< 30°C). Protect from light.

Shelf Life:

24 months

Shelf Life after Opening:

Discard any unused powder immediately after opening

Shelf Life after Dilution/Reconstitution:

24 hours

Packaging Available:

100g/packet, 100packets/carton, 10kg/carton;

500g/packet, 30packets/carton, 15kg/carton

Manufacturer and Registration Holder:

N.S. Laboratory Sdn. Bhd.

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Revised Date:

08/04/2022